Inequality and Crime: Empirical Analysis in Rio

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Abstract

Crime and inequality are critical issues in urban centers, particularly in developing countries. Rio de Janeiro, with its high levels of both economic inequality and violent crime, provides a great setting to explore the relationship between these variables at a micro level. This study investigates the mechanism through which inequality amplifies crime within the city, expanding the understanding of a dynamic already explored in the literature both theoretically and applied, mostly at city level, looking at its manifestations at a more localized scale. For that purpose, we use panel data from 2012 to 2024 on economic inequality, crime rates, and other socioeconomic controls of the 33 municipality administrative regions (A.R.) and estimate a fixed-effects model to account for unobserved heterogeneity and identify the causal impact of the relationship.

1 Introduction

Our research is motivated by the idea that exposure to inequality may facilitate the emergence of grievous criminal acts against the existing economic system and/or, as per Becker's Economic Theory of Crime, dampen opportunity costs for criminal activity.

The former motivation is linked to an extensive literature in Psychology that finds that positive feelings towards "climbing the ladder" are often conditional on climbing it higher than one's peers, while climbing it lower is associated with feelings of frustration and grievance[1]. Aversion to inequality has also long been documented in Economics, such as in the well-known experimental results of the Ultimatum Game.

The city of Rio de Janeiro has traditionally been a poster child of Brazil's extreme inequality rates. Favelas have historically been side-by-side to the country's most affluent neighborhoods. We thus hypothesize that Rio's unique development has had a marked impact in shaping the city's historical and current criminal rates.

2 Data

To evaluate the data within the city of Rio de Janeiro, we used the division of its 33 Administrative Regions in each one of the sources and variables. The description of each A.R. and its neighborhood reach is available in the Appendix table. The sample covers the period from 2012 to 2024.

2.1 Inequality

The inequality data in this study comes from IBGE's (Brazilian Institute of Geography and Statistics) PNADc (National Household Sample Survey - Continuous), which provides sensitive socioeconomic data at the household level, aggregated here at the A.R. level.

To measure inequality, we compute yearly Gini Indices for each A.R.

2.2 Criminality

The criminality data used here covers different crime indicators, such as violent crimes, property crimes, imprisonment, etc.

We obtained data regarding all crimes formally accounted by the police from Instituto de Segurança Pública (ISP). The lowest hierarchical levels are those of CISPs and UPPs.

From Instituto Fogo Cruzado (IFG), we obtained data on shootings. Each shooting observation contains geocoordinates, datetime, information on civil/police letality and fatalities etc.

2.3 Controls

In order to control for other variables that might also affect crime, we also collected data on average income, poverty, education, unemployment, population, area density and urbanization, police presence, public security policies, favela presence, faction presence, victimization.

From PNADc, we extracted data relating to average income, unemployment rate, poverty, urbanization and favela presence. The poverty index is constructed as the percentage of the A.R. population with income below the poverty line.

IFG's datasets also allow us to assess variables such as police presence and *chacina* occurrences during shootings. In partnership with GENI/UFF, IGF has also published a geopolitical map of the city (see here), from 2008 to 2023, where the areas controlled by facctions and the militia are identified.

We propose using IVs for police presence and public security to account for endogeneity (as investment on public security might itself be determined by inequality as well). Barros and Vries (2017) use party alignment between the governor and mayors in the state of São Paulo.

[1] See, for example, Yu and Wang (2017) and Gao et al. (2022).

3 Methodology

3.1 Base Model

The base model consists in a fixed-effects regression model:

$$Y_{it} = \beta_0 + \beta_i X_{it} + \lambda_i I_{it} + \gamma_t + \gamma_i + \epsilon_{it}$$

- Y_{it}: criminality
- X_{it}: vector of control variables
- I_{it}: inequality index

3.2 Model extensions

- Crime inertia: painel GMM (lagged vbs)
- Correlation between ind var: ?????
- Causality inversion: ?????
- Propagation between A.R.: ?????

4 Results

TBA

References

[1] Lucas P. G. de Barros and Luiz Felipe de Vries. "The Effects of Inequality on Crime: A CrossSectional Analysis". In: *Nome do periódico* (2017).

[2] Lingxi Gao et al. "How Wealth Inequality Affects Happiness: The Perspective of Social Comparison". In: *Frontiers in Psychology* (2022). DOI: ⟨https://doi.org/10.3389/fpsyg.2022.829707⟩.

[3] Zonghuo Yu and Fei Wang. "Income Inequality and Happiness: An Inverted U-Shaped Curve". In: *Frontiers in Psychology* (2017). DOI: ⟨https://doi.org/10.3389/fpsyg.2017.02052⟩.

Appendix

A. R.	Coverage Area
Portuária	Caju, Gamboa, Saúde, Santo Cristo
Centro	Centro
Rio Comprido	Catumbi, Estácio, Cidade Nova, Rio Comprido
Botafogo	Botafogo, Catete, Cosme Velho, Flamengo, Glória, Humaitá, Laranjeiras, Urca
Copacabana	Copacabana, Leme
Lagoa	Gávea, Ipanema, Jardim Botânico, Lagoa, Leblon, São Conrado, Vidigal
São Cristóvão	Benfica, Mangueira, São Cristóvão, Barreira do Vasco
Tijuca	Tijuca, Praça da Bandeira, Alto da Boa Vista
Vila Isabel	Vila Isabel, Andaraí,Grajaú, Maracanã
Ramos	Ramos, Bonsucesso, Olaria
Penha	Penha, Penha Circular, Brás de Pina
Inhauma	Inhaúma, Engenho da Rainha, Higienópolis, Tomás Coelho
Méier	Méier, Abolição, Água Santa, Engenho de Dentro, Engenho Novo, Jacaré, Lins de Vasconcelos, Pilares, Riachuelo, Rocha, Sampaio, São Francisco Xavier, Todos os Santos, Piedade, Encantado
Irajá	Irajá, Colégio, Vicente de Carvalho, Vila Cosmos, Vila da Penha, Vista Alegre
Madureira	Madureira, Bento Ribeiro, Campinho, Cascadura, Honório Gurgel, Marechal Hermes, Oswaldo Cruz, Rocha Miranda, Vaz Lobo, Turiaçú
Jacarepaguá	Jacarepaguá, Anil, Curicica, Freguesia, Tanque, Gardênia Azul, Jacarepaguá, Pechincha, Praça Seca, Taquara, Vila Valqueire
Bangu	Bangu, Gericinó
Campo Grande	Campo Grande, Santíssimo, Senador Vasconcelos
Santa Cruz	Santa Cruz
Ilha do Governador	Bancários, Cacuia, Cidade Universitária, Cocotá, Freguesia, Galeão, Jardim Carioca, J. Guanabara, Moneró, Pitangueiras, Portuguesa, Praia da Bandeira, Tauá, Ribeira, Zumbi
Paquetá	Paquetá
Anchieta	Anchieta, Guadalupe, Parque Anchieta, Ricardo de Albuquerque
Santa Teresa	Santa Teresa
Barra da Tijuca	Barra da Tijuca, Itanhangá, Joá
Pavuna	Pavuna, Barros Filho, Costa Barros
Guaratiba	Guaratiba, Barra de Guaratiba, Pedra de Guaratiba
Rocinha	Rocinha
Jacarezinho	Jacarezinho
Complexo do Alemão	Complexo do Alemão
Complexo da Maré	Complexo da Maré
Vigário Geral	Vigário Geral, Cordovil, Parada de Lucas, Jardim América
Realengo	Realengo, Deodoro, Vila Militar, Campo dos Afonsos, Jardim Sulacap, Magalhães Bastos
Cidade de Deus	Cidade de Deus

Table 1: Administrative Regions of Rio de Janeiro